AQ8.5: Activity Questions 5 - Not Graded

This assignment will not be graded and is only for practice.
Level 1

1 point
Choose the correct options.

- [ ] Let $T: \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation, where $\mathbb{R}^2$ is the inner product space with respect to the dot product. Then $Tu \cdot Tv = u \cdot v$.
- [ ] Let $T: \mathbb{R}^2 \to \mathbb{R}^2$ be an orthogonal linear transformation, where $\mathbb{R}^2$ is the inner product space with respect to the dot product. Then $Tu \cdot Tv = u \cdot v$.
- [ ] Let $T: \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation, where $\mathbb{R}^2$ is the inner product space with respect to the inner product given by $< a, b > = 2a_1b_1 + 5a_2b_2$. Then $\langle Tu, Tv \rangle = \langle u, v \rangle$.
- [ ] Let $T: \mathbb{R}^2 \to \mathbb{R}^2$ be an orthogonal linear transformation, where $\mathbb{R}^2$ is the inner product space with respect to the inner product given by $< a, b > = 2a_1b_1 + 5a_2b_2$. Then $\langle Tu, Tv \rangle = \langle u, v \rangle$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Let $T: \mathbb{R}^2 \to \mathbb{R}^2$ be an orthogonal linear transformation, where $\mathbb{R}^2$ is the inner product space with respect to the dot product. Then $Tu \cdot Tv = u \cdot v$.
Let $T: \mathbb{R}^2 \to \mathbb{R}^2$ be an orthogonal linear transformation, where $\mathbb{R}^2$ is the inner product space with respect to the inner product given by $< a, b > = 2a_1b_1 + 5a_2b_2$. Then $\langle Tu, Tv \rangle = \langle u, v \rangle$.

1 point
If A is the matrix representation of an orthogonal transformation $T: \mathbb{R}^n \to \mathbb{R}^n$, then choose the set of correct statements.

- [ ] $AA^T = I$
- [ ] $A = A^T$
- [ ] AA^T is an upper triangular matrix
- [ ] AA^T is a lower triangular matrix
- [ ] $A^T = A^{-1}$
- [ ] $A^T = -A^{-1}$
- [ ] A^{-1} does not exist
- [ ] A^{-1} is also an orthogonal matrix

No, the answer is incorrect.
Score: 0

Accepted Answers:
$AA^T = I$
AA^T is an upper triangular matrix
AA^T is a lower triangular matrix
$A^T = A^{-1}$
A^{-1} is also an orthogonal matrix

1 point

If A is the matrix representation of an orthogonal transformation $T: R^n \rightarrow R^n$, then choose the set of correct statements.

☐ The column vectors of A are orthogonal but not orthonormal.

☐ The column vectors of A are orthonormal.

☐ The row vectors of A are orthogonal but not orthonormal.

☐ The row vectors of A are orthonormal.

No, the answer is incorrect.
Score: 0
Accepted Answers:
The column vectors of A are orthonormal.
The row vectors of A are orthonormal.

Level 2

1 point

Consider the orthogonal set $B = \{(1,1,1), (-1,1,0)\}$ in R^3. Find a third vector in R^3 which is orthogonal to both the vectors in B.

○ $(1,1,-2)$

○ $(1,-1,2)$

○ $(-1,1,-2)$

○ $(-1,-1,-2)$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$(1,1,-2)$

1 point

Consider the orthogonal set $B = \{(1,1,1), (-1,1,0)\}$ and vector obtained in question 4 which is orthogonal to the set B. Form the columns of an orthogonal matrix U by scaling these vectors appropriately. Choose the correct option.

○ $U = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & -2 \end{bmatrix}$

○

$U = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{-2}{\sqrt{6}} \end{bmatrix}$ U = [3

√

1 3

√

1 3

√

1 2

√

−1 2

√

10 6

√

1 6

√

1 6

√

−2]

○ $U = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 1 & 1 \\ 1 & 0 & -2 \end{bmatrix}$ U = [111−110−11−2]

○ $U = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{-1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{-2}{\sqrt{6}} \end{bmatrix}$ U = [3

√

1 3

√

1 3

√

1 2

√

−1 2

√

10 6

√

−1 6

√

1 6

√

−2]

No, the answer is incorrect.
Score: 0
Accepted Answers:

$U = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{-2}{\sqrt{6}} \end{bmatrix}$U = [3

√
1 3
√
1 3
√
1 2
√
−1 2
√
10 6
√
1 6
√
1 6
√

−2]

1 point

Consider a vector $v = (3, 0, 4)$v = (3, 0, 4) and UU from Question 5. Choose the correct option using the dot product as the standard inner product.

- $\| v \| = \| Uv \|$ ‖v‖ = ‖Uv‖
- $\| v \| \neq \| Uv \|$ ‖v‖ = ‖Uv‖
- $2 \| v \| = \| Uv \|$ 2‖v‖ = ‖Uv‖
- $\| v \| = 2 \| Uv \|$ ‖v‖ = 2‖Uv‖

No, the answer is incorrect.
Score: 0
Accepted Answers:

$\| v \| = \| Uv \|$ ‖v‖ = ‖Uv‖

1 point

Let the matrix representation of a linear transformation $T: R^3 \rightarrow R^3$ T : R3 → R3 be $U = \begin{bmatrix} \frac{-1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix}$ U =

[2

√
−1 2
√
10 6
√
−1 6
√
−1 6
√
2 3
√
1 3
√
1 3
√
1] .

Let $B = \begin{bmatrix} 1 & 3 & 3 \\ 3 & 1 & 3 \\ 3 & 3 & 1 \end{bmatrix}$ B = [133313331] . Choose the set of correct statements.

- [ ] T T is an orthogonal transformation.
- [ ] T T is not an orthogonal transformation.
- [ ] B B is a real symmetric matrix.
- [ ] $U^{-1}BU$ U−1BU is a diagonal matrix.
- [ ] $U^{-1}BU$ U−1BU is a scalar matrix.
- [ ] $U^{-1}BU$ U−1BU is an upper triangular matrix.
- [ ] $U^{-1}BU$ U−1BU is a lower triangular matrix.

No, the answer is incorrect.
Score: 0
Accepted Answers:
T T is an orthogonal transformation.
B B is a real symmetric matrix.
$U^{-1}BU$ U−1BU is a diagonal matrix.